LAMIA ISLAM

AI SAFETY · HUMAN-CENTRED AI · ROBOTICS BANGLADESH

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ICE graduate from Pabna University of Science and Technology working on AI safety, human-centred AI and robotics. I build AI systems that recognise when they may be wrong about a person, act with care when they are, and learn from the mistake. My work spans wearable physiological sensing, out-of-distribution detection, and safe, human-overridable intervention in robots and autonomous systems, with four peer-reviewed conference papers and two journal manuscripts under review.

EDUCATION

Pabna University of Science and Technology (PUST) Feb 2022 – Aug 2026
B.Sc. in Information and Communication Engineering

- Thesis: *Design and Evaluation of a Competence-Gated Driver-State Monitoring System for Autonomous Intervention Control*. Supervisor: Dr. Md. Anwar Hossain. Defended 2 September 2026.

RESEARCH EXPERIENCE

Research Assistant, Department of ICE, PUST 2026 – Present
Supervisor: Dr. Md. Anwar Hossain

- Built a density-based classifier of physiological state (stress) from ECG and electrodermal activity, trained on WESAD (15 laboratory participants) and evaluated leave-one-subject-out.
- Tested it on real driving recordings (MIT DriveDB) from people it had never seen. Showed that confidence scores rank these unfamiliar people worse than chance, while the model's marginal log-evidence, a value it already computes and normally discards, detects them at no extra cost.
- Showed that how well a detector ranks unfamiliar data does not predict how it behaves as a gate on intervention, which is what a person actually experiences.
- Found participants poorly described by the model even within its own training data, and caught a scale artifact between datasets that inflated detection before reporting.
- Deployed the monitor in ROS 2 Humble and Gazebo as a graded, human-overridable intervention (alert, slow, stop). Further experiments running; manuscript in preparation.

Research Assistant, Department of ICE, PUST 2024 – Mar 2026
Supervisor: Dr. Md. Imran Hossain

- Carried three projects from formulation to publication: nightmare prediction in children, hepatitis C detection, and an RFID access-control system with networked logging [C1, C2, C4].

JOURNAL MANUSCRIPTS UNDER REVIEW

Site identity leaks through missingness: unsupervised competence monitoring under-detects shift in four cardiovascular cohorts
L. Islam *(first and corresponding author), M. Einollahi Asgarabad | Royal Society Open Science | Submitted Oct 2026*

- Asked whether a clinical model can tell when patients come from a site it was never trained on, using the four separate cohorts of the UCI Heart Disease collection (851 records, leave-one-cohort-out).
- Unsupervised monitors missed most of the real shift between sites, even though a supervised classifier could detect it on the same data; the sites could also be told apart from missing-value patterns alone.
- Audited widely shared copies of the dataset and found duplicated and mislabelled records that published work still relies on.
- Wrote all analysis code. Code and audit: doi.org/10.5281/zenodo.23205549

Vision-based crack segmentation and coverage planning for autonomous mobile robots
M. B. A. Z. Shammo, T. Debnath, ***L. Islam****, S. Sarker, J. Habib | Int. J. of Intelligent Robotics and Applications (Springer) | Revision submitted Sep 2026*

- An inspection robot that detects cracks with a lightweight segmentation model and plans its own coverage of the area, avoiding obstacles with infrared sensing.
- Preprint: researchsquare.com/article/rs-9996410

PEER-REVIEWED CONFERENCE PAPERS

[C1] **L. Islam**, M. B. A. Z. Shammo, M. I. Hossain. Real-time RFID authentication and access control system with networked logging. *IEEE QPAIN*, Chattogram, Apr 2026. doi:10.1109/QPAIN69676.2026.11546467

[C2] M. B. A. Z. Shammo, **L. Islam**, M. Hasan, M. A. Hossain, M. I. Hossain, M. A. Mia, M. H. Mohona. Machine learning-based hepatitis C detection using ensemble boosting. *3rd Int. Conf. on Big Data, IoT and Machine Learning*, Springer Nature, Apr 2026. doi:10.1007/978-3-032-15764-5_50

[C3] M. A. Miya, M. H. Mohona, M. A. Hossain, M. B. A. Z. Shammo, A. Islam, **L. Islam**. Remote health monitoring via PPG signal processing and machine learning using mobile devices. *IEEE ICCIT*, Cox's Bazar, Dec 2025. doi:10.1109/ICCIT68739.2025.11491546

[C4] **L. Islam**, M. B. A. Z. Shammo, M. Rashed, M. I. Hossain, M. A. Miya, M. H. Mohona. Machine learning-based prediction of nightmares in children: a pediatric psychological integration. *IEEE COMPAS*, Kushtia, 2025. doi:10.1109/COMPAS67506.2025.11381639

RESEARCH PROJECTS

Multi-Robot Search and Rescue 2026

ROS 2 Humble, Gazebo Fortress | Submitted to IEEE ICCIT 2026

- Three robots map a collapsed building together and share a belief map of where survivors are likely to be, using SLAM, map merging and frontier assignment.
- Found that this belief helps in open space but misleads robots in rubble, because it spreads through walls; a path-aware (geodesic) version partly corrects it.
- Code: github.com/Lamia-Islam

When Benchmarks Mislead: Wound Segmentation 2026

22 methods, synthetic and clinical datasets | Manuscript in preparation

- Two synthetic benchmarks ranked the same methods in opposite orders while two clinical datasets agreed; background texture decides whether a synthetic benchmark can be trusted.
- Preprint: doi.org/10.21203/rs.3.rs-10362132/v1

SKILLS

Machine Learning Python, scikit-learn, PyTorch, TensorFlow; out-of-distribution detection, calibration, leave-one-subject-out and leave-one-cohort-out validation, statistical testing

Robotics ROS 2 Humble, Gazebo Fortress, slam_toolbox, MoveIt 2, RViz, Ubuntu (WSL2)

Signals and Hardware ECG, EDA and PPG processing, HRV features; ESP32, AD8232, GSR sensors, Arduino, RFID

Other C, R, MATLAB, Git, LaTeX

SERVICE AND LEADERSHIP

Robotics Secretary, ICE Association, PUST Oct 2025 – Present

Peer Reviewer, IEEE Conferences

IEEE ICCIT 2026; IEEE RAAICON 2026 (8 papers); IEEE RAAICON 2025

REFERENCES

Dr. Md. Anwar Hossain

Professor and Chairman, Department of ICE, and Dean, Faculty of Engineering and Technology, PUST. Thesis supervisor.

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Dr. Md. Imran Hossain

Associate Professor, Department of ICE, PUST. Research mentor, 2024 to 2026.

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